Enhancing Syd's Health Platform with Contactless BP, SpO_2, and BMI Estimation

Group 2: Ziyi, Jiale, Samson, Liza

Client: iamYiam – Syd Platform

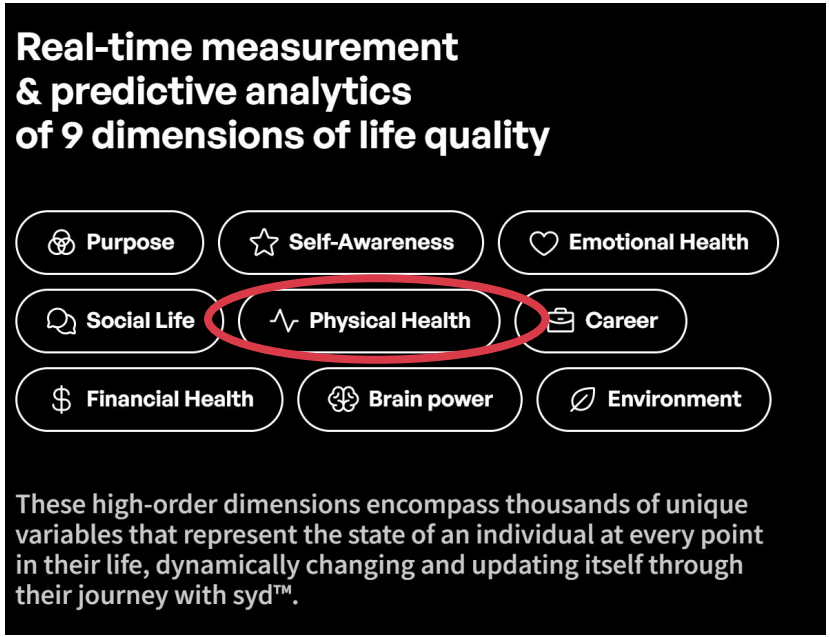
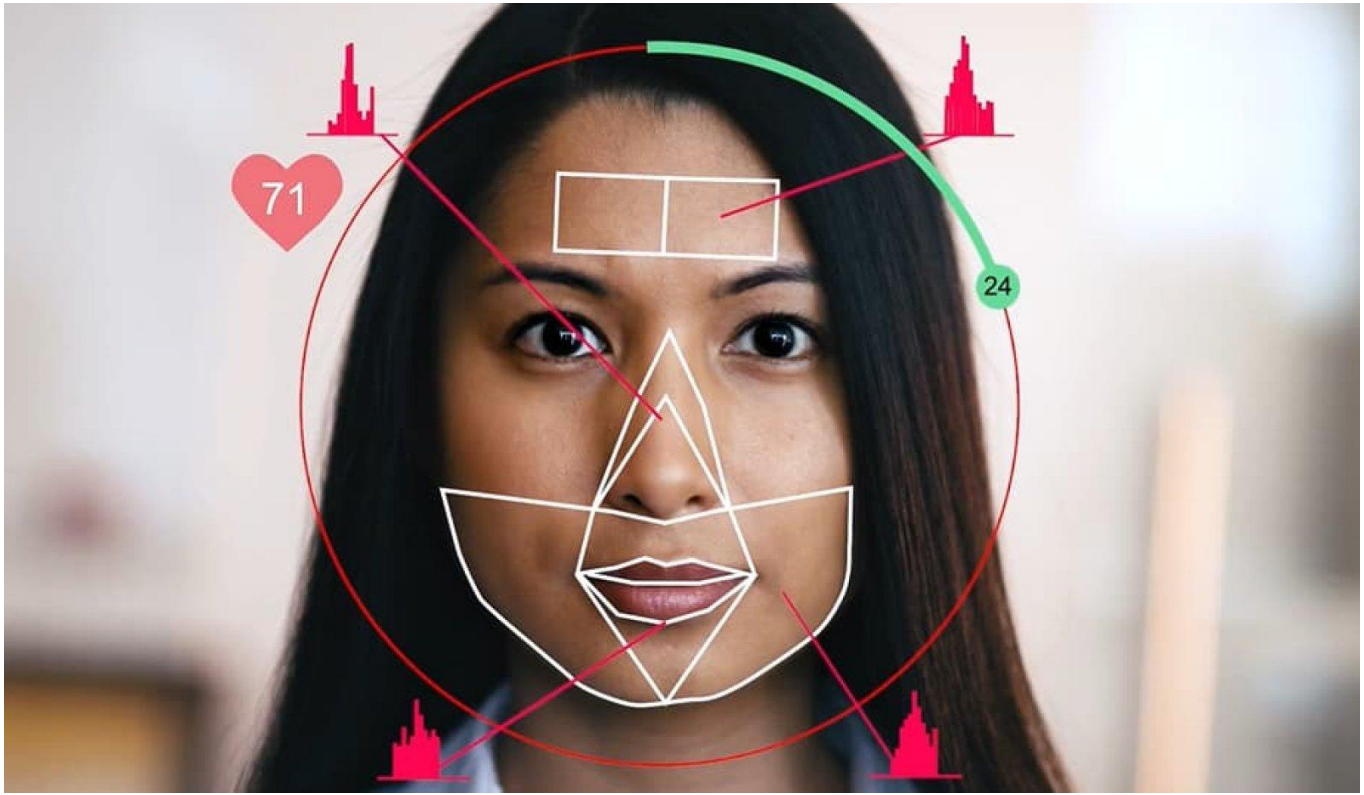


Image Source: https://syd.life

BG [2/3]



Blood Pressure

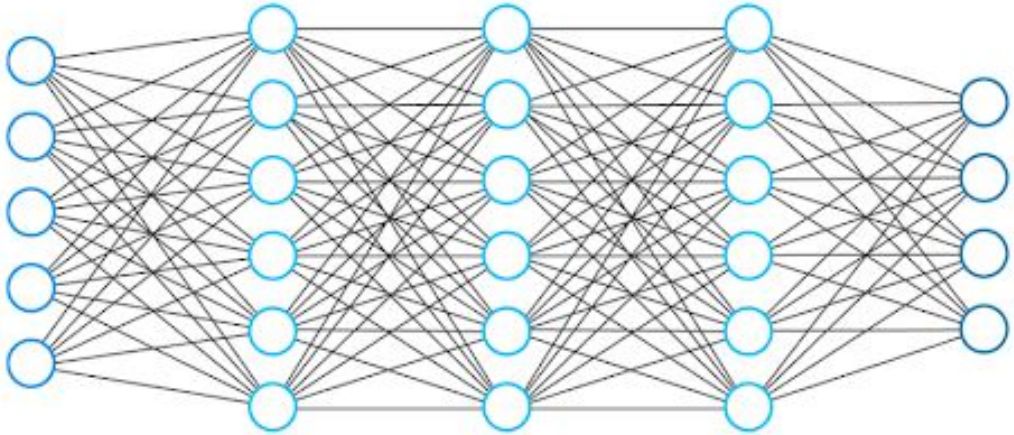
Blood Oxygen

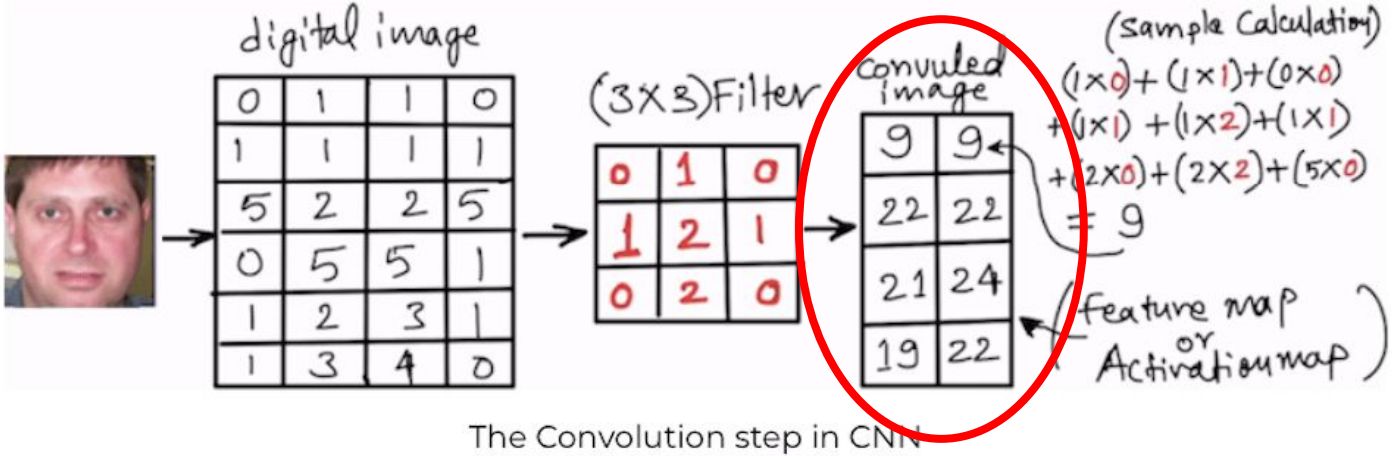
BMI

Health Biomarker	Required Accuracy/Validation	Relevance to Underwriting
Blood Pressure (SBP/DBP)	MAE: **~3**-5% (SBP), ~5-**7%** (DBP) in real-world tests [1]	**High:** Core vital sign for mortality/morbidity risk. Direct input for risk classification.
Blood Oxygen (SpO_2)	MAE: 1.274%; RMSE: **1.71%** (exceeds medical standard) [2]	**High:** Key indicator of respiratory and cardiac function.

[1] https://www.researchgate.net/publication/378264644_Robust_blood_pressure_measurement_from_facial_videos_in_diverse_environments
[2] https://pmc.ncbi.nlm.nih.gov/articles/PMC10968547/

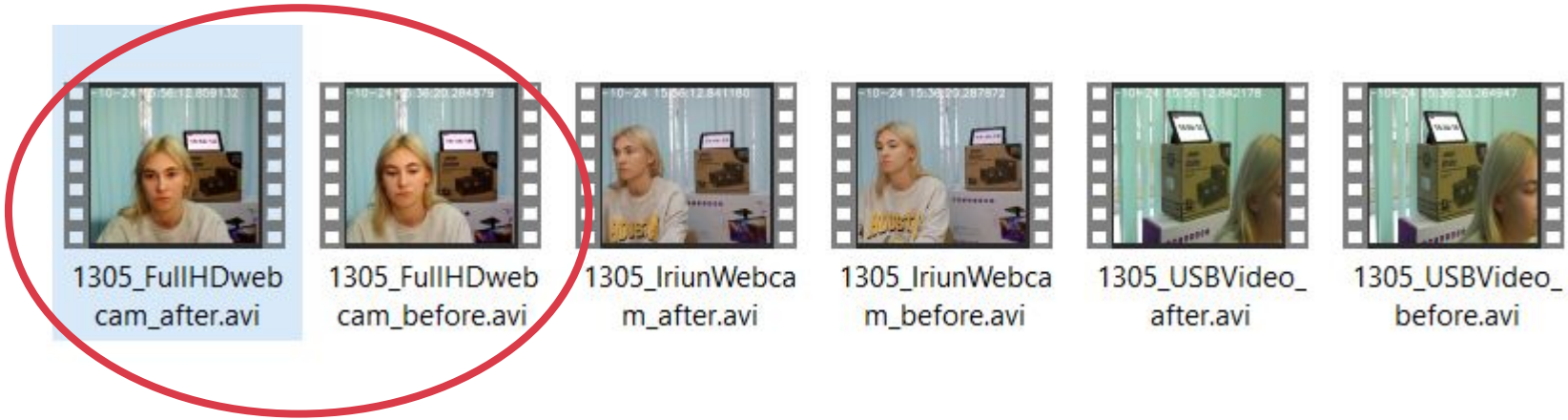
Deep Learning





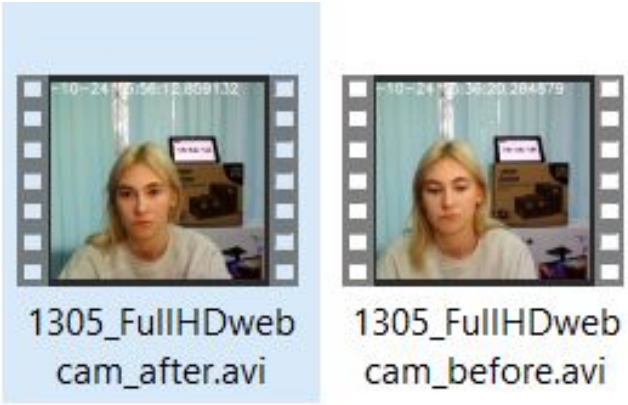
The Convolution step in CNN

MCD-rPPG Dataset



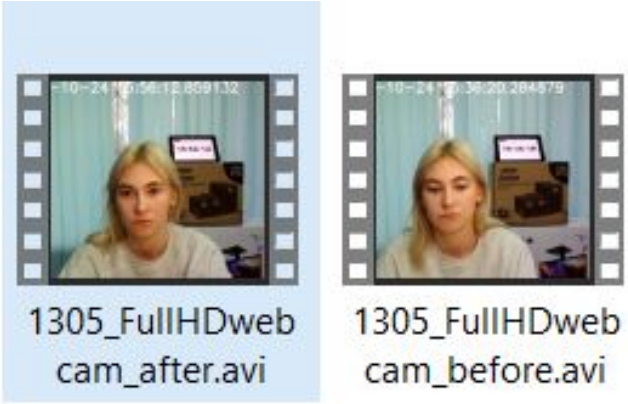
- 600 subjects × 2 states × 3 cameras
- 1,200 videos — 3 min each

MCD-rPPG Dataset



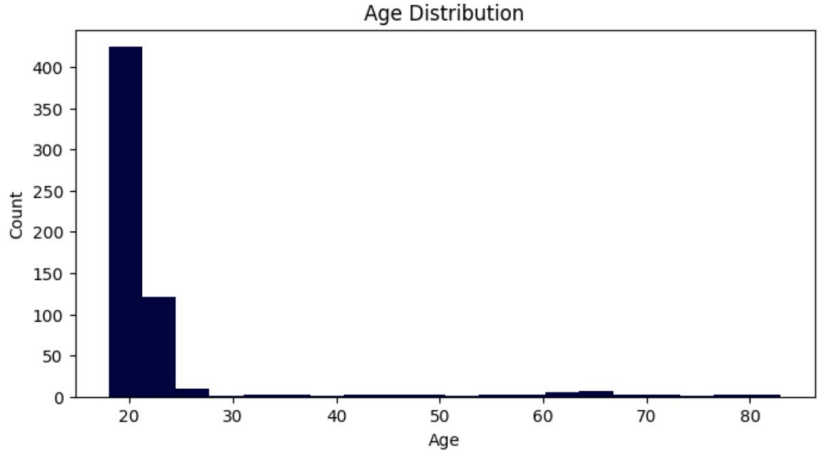
- 600 subjects × 2 states
- 1,200 videos — 3 min each

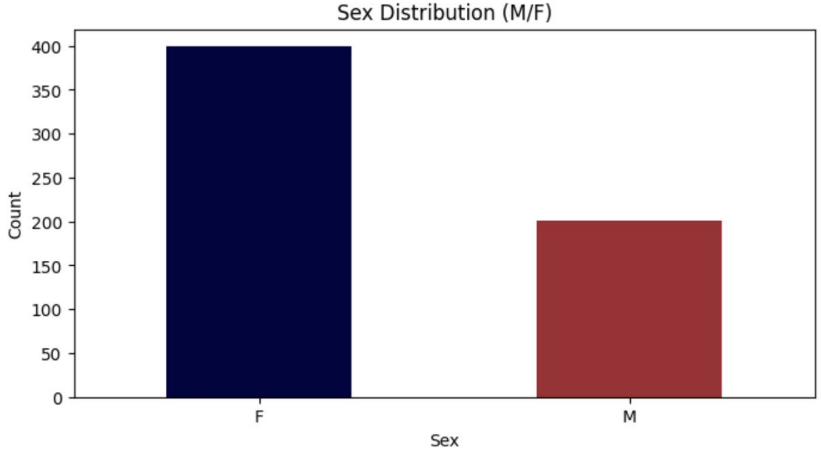
MCD-rPPG Dataset



- 600 subjects × 2 states
- 1,200 videos — 3 min each

```
Age Statistics:
Mean: 23.08, Std: 10.90, Min: 18.0, Max: 83.0
```





patient_id	bmi	upper_ap	lower_ap	saturation	step	camera	view	
1020	19.031141868512112	113.0	78.0	98.0	after	FullHDwebcam	front	vide
1020	19.031141868512112	105.0	78.0	99.0	before	FullHDwebcam	front	vide
1024	17.056176513282043	115.0	69.0	99.0	after	FullHDwebcam	front	vide
1024	17.056176513282043	106.0	59.0	99.0	before	FullHDwebcam	front	vide
1035	21.366869143188545	112.0	72.0	98.0	after	FullHDwebcam	front	vide
1035	21.366869143188545	106.0	55.0	98.0	before	FullHDwebcam	front	vide
1091	27.60945175517229	138.0	87.0	98.0	after	FullHDwebcam	front	vide
1091	27.60945175517229	135.0	86.0	98.0	before	FullHDwebcam	front	vide
1097	24.609733700642792	116.0	71.0	98.0	after	FullHDwebcam	front	vide
1097	24.609733700642792	112.0	70.0	98.0	before	FullHDwebcam	front	vide

DS [2/3]

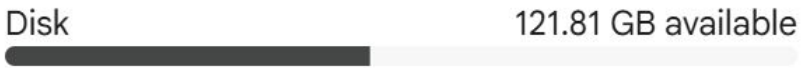
video	upper_ap	lower_ap	saturation	bmi
video/1020_FullHDwebcam_after.avi	113.0	78.0	98.0	19.031141868512112
video/1020_FullHDwebcam_before.avi	105.0	78.0	99.0	19.031141868512112
video/1024_FullHDwebcam_after.avi	115.0	69.0	99.0	17.056176513282043
video/1024_FullHDwebcam_before.avi	106.0	59.0	99.0	17.056176513282043
video/1035_FullHDwebcam_after.avi	112.0	72.0	98.0	21.366869143188545
video/1035_FullHDwebcam_before.avi	106.0	55.0	98.0	21.366869143188545
video/1091_FullHDwebcam_after.avi	138.0	87.0	98.0	27.60945175517229
video/1091_FullHDwebcam_before.avi	135.0	86.0	98.0	27.60945175517229
video/1097_FullHDwebcam_after.avi	116.0	71.0	98.0	24.609733700642792
video/1097_FullHDwebcam_before.avi	112.0	70.0	98.0	24.609733700642792

Large Dataset vs. Computing Resources

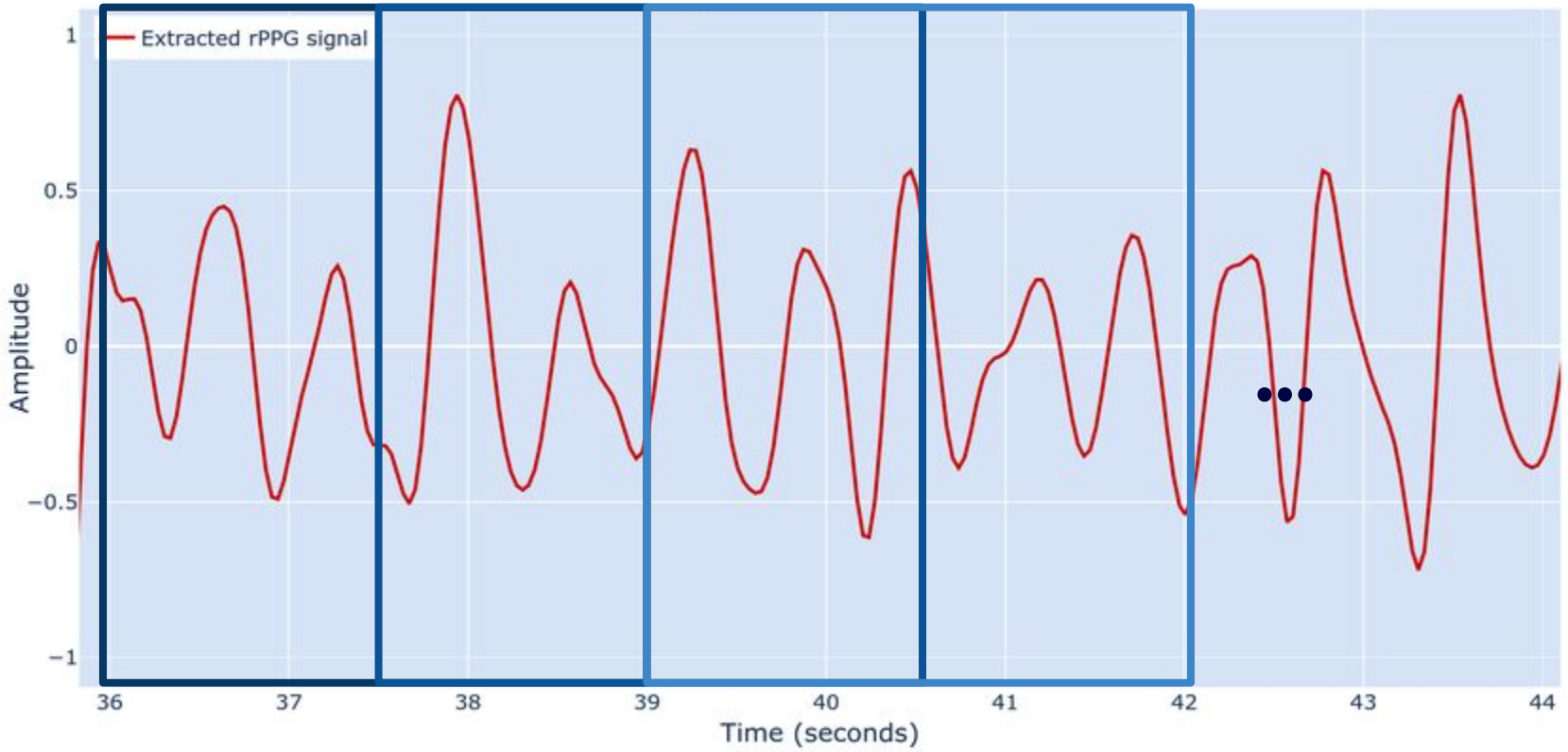
Dataset size: 50 GB total

- Split into Train/Validation/Test

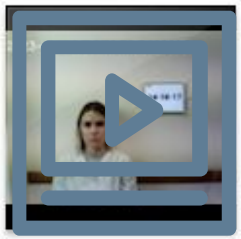




General Processing



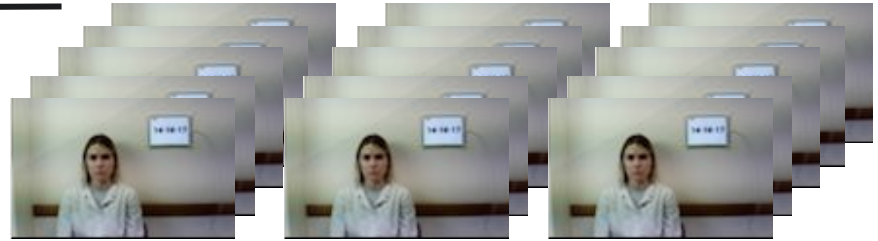
Data Preparation



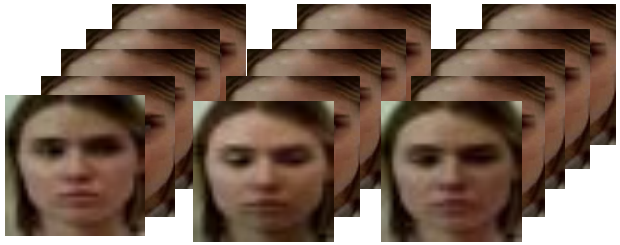
Extract Frames



Segmentation



Seg. 1 Seg. 2 Seg. 3

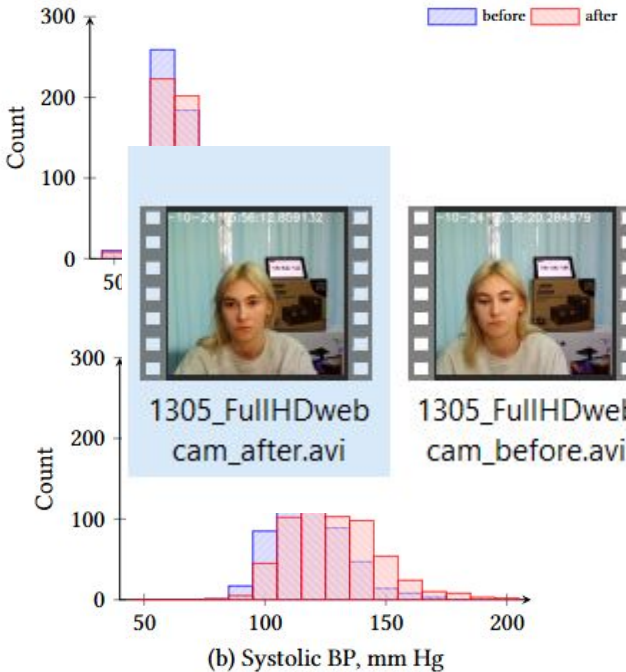


Face Detection



Table 2: Distribution of anthropometric parameters and biomarkers in the dataset

	mean	std	min	max
Weight, kg	65.92	15.79	43.00	168.00
Height, cm	169.75	8.87	147.00	201.00
BMI, kg/m^2	22.73	4.34	15.39	47.03
Age, years	23.08	10.90	18.00	83.00



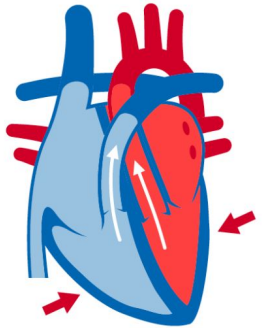


Progress on models - bp

Blood Pressure

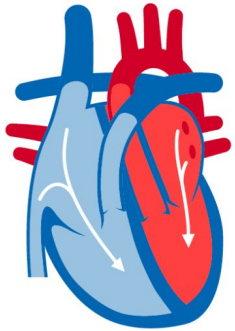
120

SBP



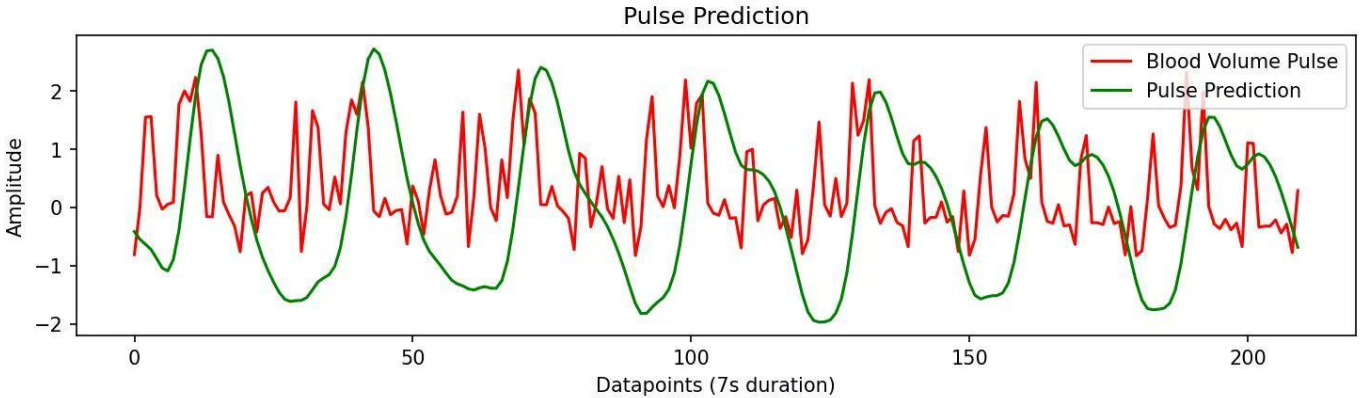
80

DBP



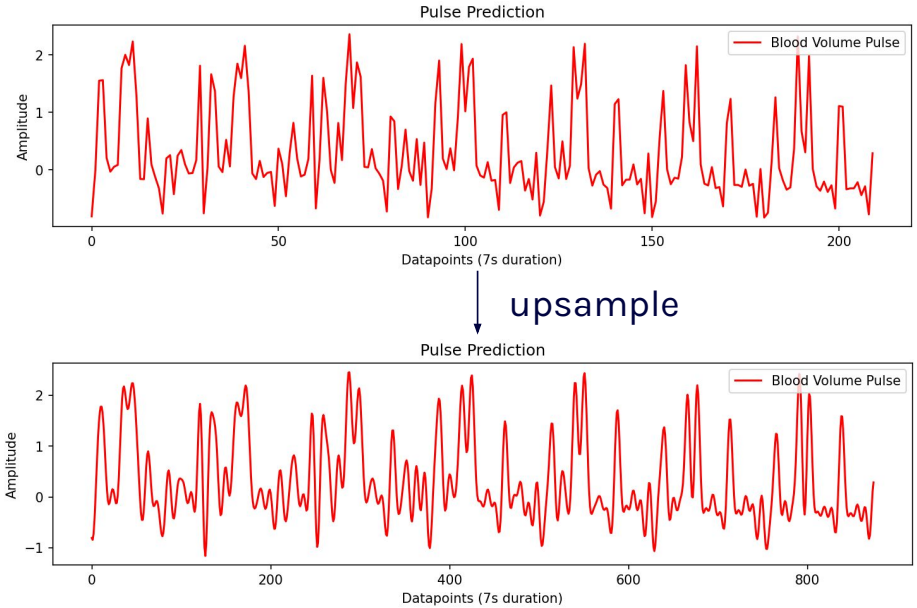
Data Extraction



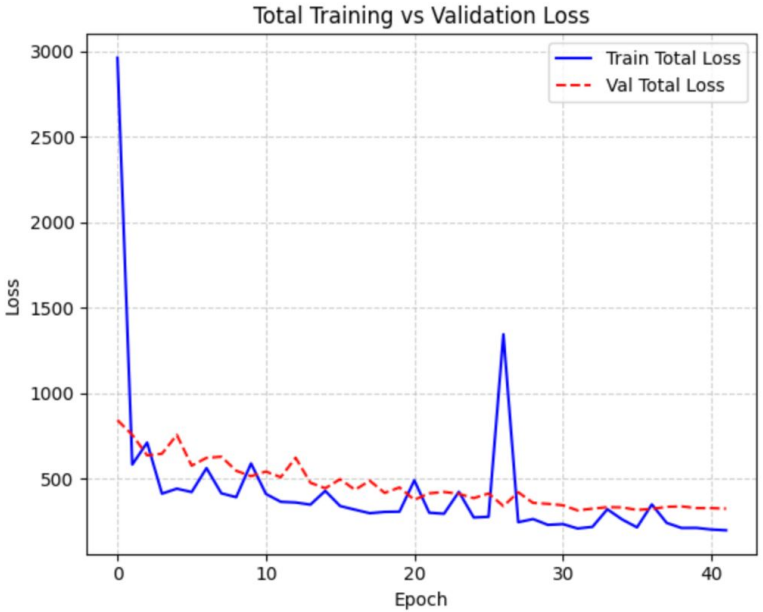


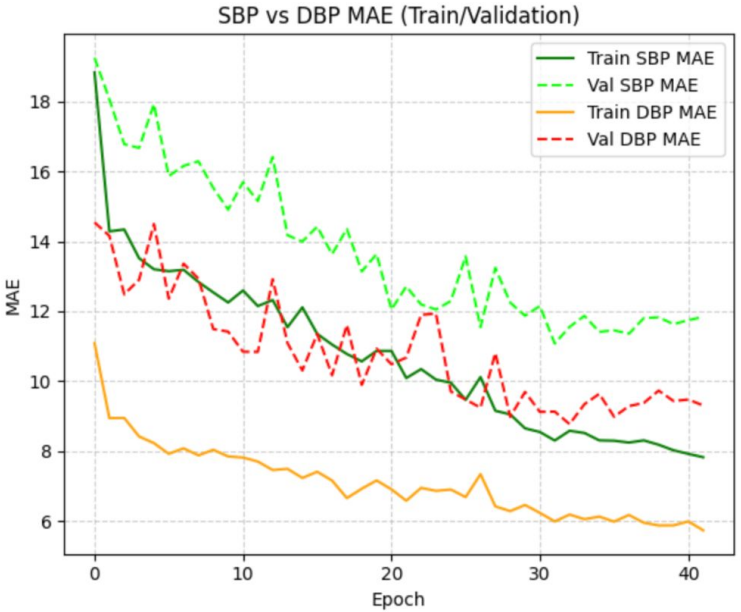
Model Selection

Blood Volume Pulse → Upsample → BP Prediction Model: **AlexNet** or **ResNet** or **LSTM** → SBP, DBP

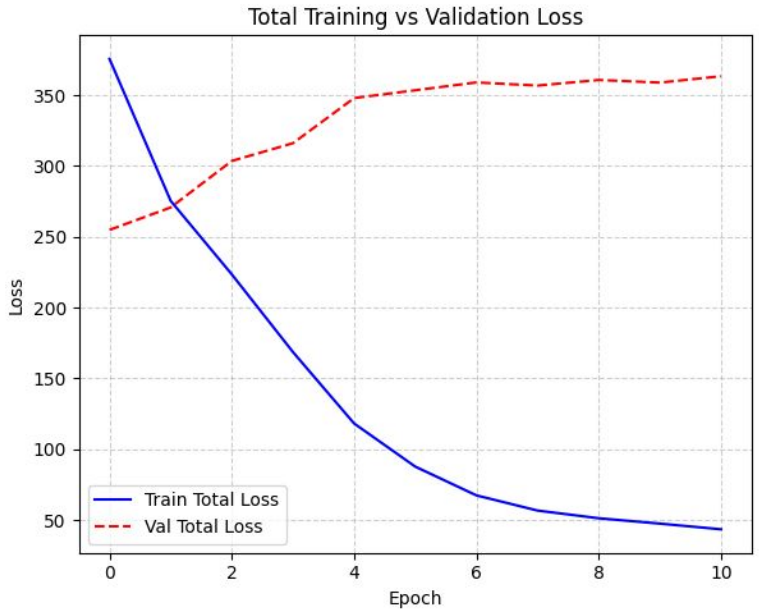


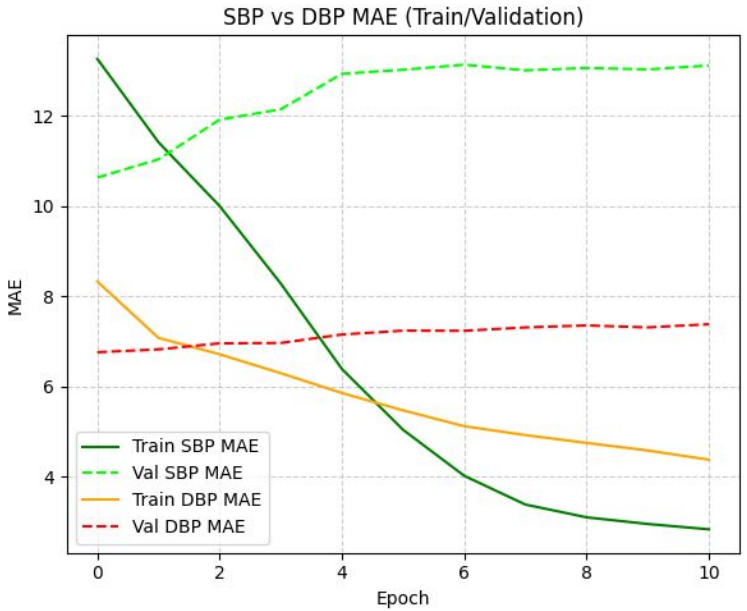
Fine Tune - AlexNet



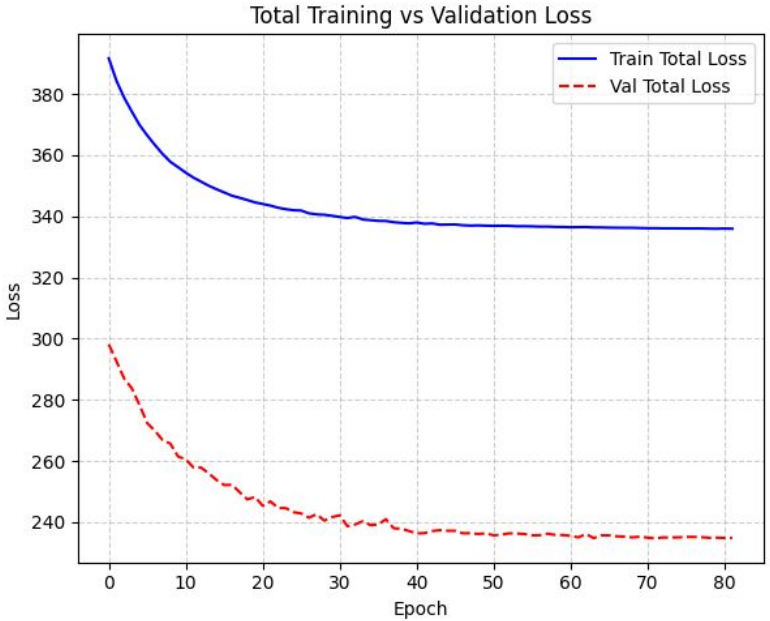


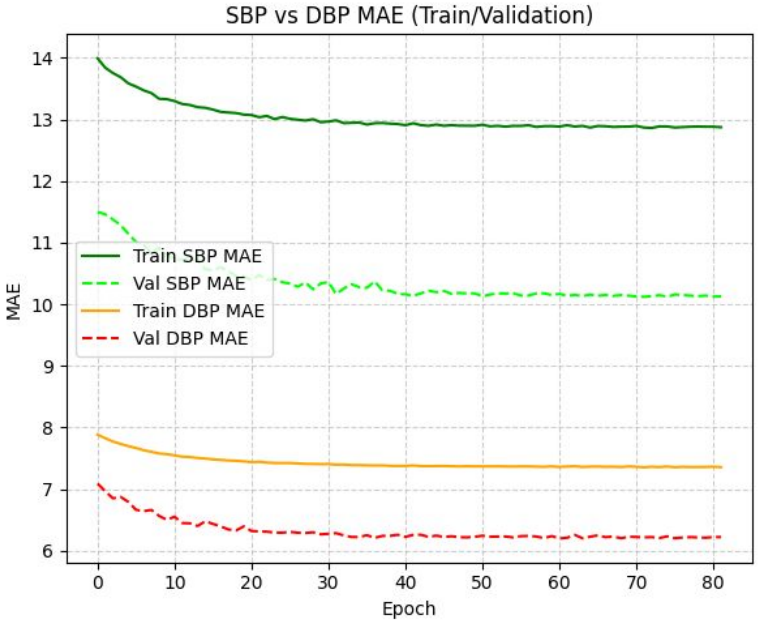
Fine Tune - ResNet



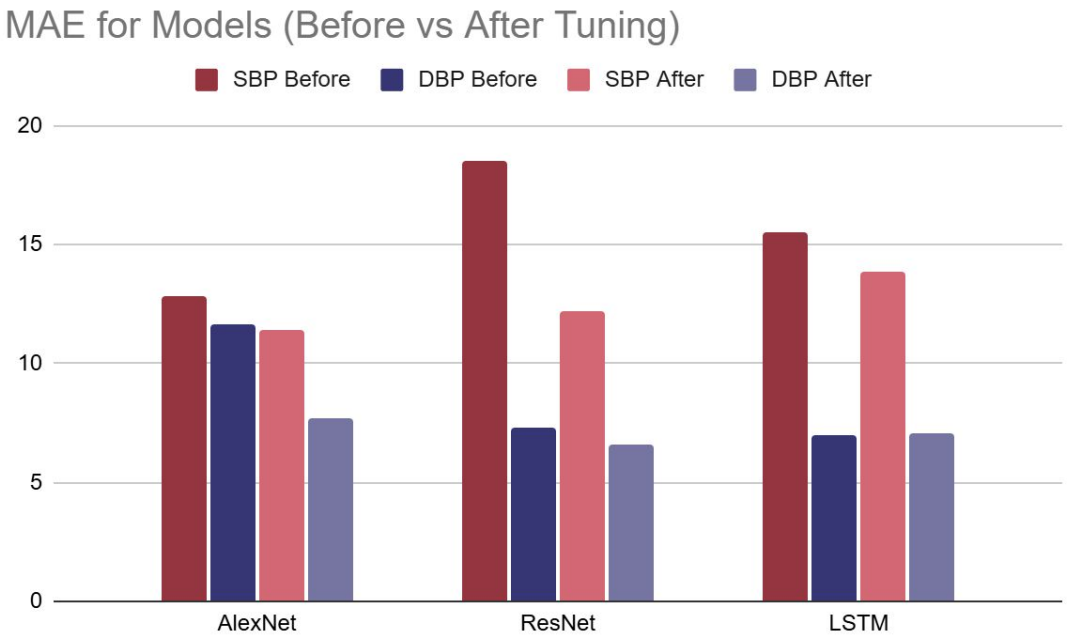


Fine Tune - LSTM





Results Comparison



11.40
SBP

6.63
DBP

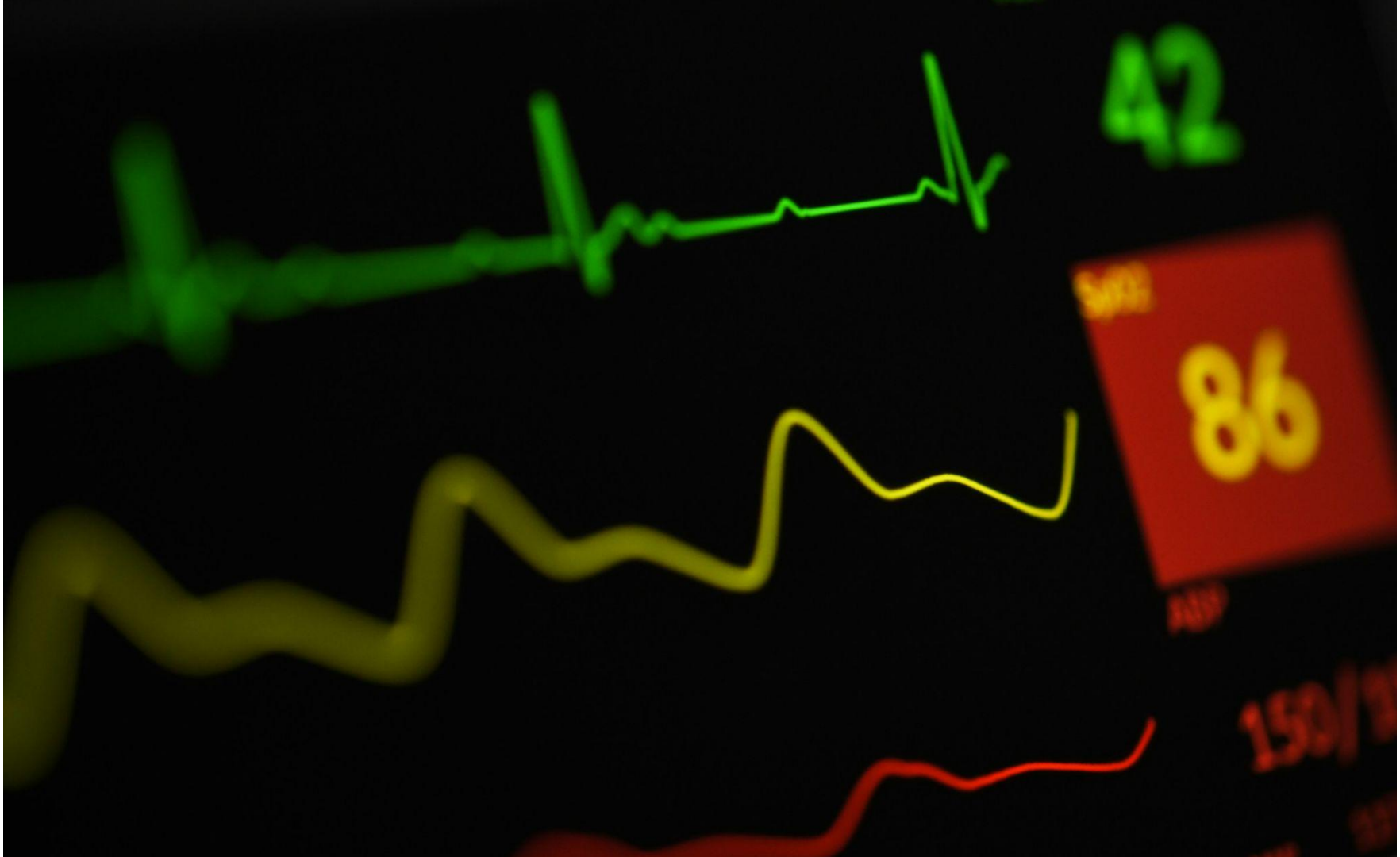
Lowest MAE

Model Comparison

Model	Before Tuning		After Tuning	
	SBP	DBP	SBP	DBP
AlexNet	avg_MAE=**12.85** avg_MAPE=11.02% median_MAE=12.71 median_MAPE=10.94%	avg_MAE=11.64 avg_MAPE=14.75% median_MAE=11.90 median_MAPE=15.08%	avg_MAE=**11.40** avg_MAPE=9.66% median_MAE=11.47 median_MAPE=9.97%	avg_MAE=7.68 avg_MAPE=10.07% median_MAPE=10.13% median_MAE=7.57
ResNet	avg_MAE=18.49 avg_MAPE=17.16% median_MAE=19.02 median_MAPE=17.68%	avg_MAE=7.27 avg_MAPE=10.10% median_MAE=7.02 median_MAPE=9.88%	avg_MAE=12.22 avg_MAPE=10.78% median_MAE=12.19 median_MAPE=10.76%	avg_MAE=**6.63** avg_MAPE=9.47% median_MAE=6.61 median_MAPE=9.45%
LSTM	avg_MAE=15.54 avg_MAPE=13.93% median_MAE=15.53 median_MAPE=13.93%	avg_MAE=**6.99** avg_MAPE=9.79% median_MAE=6.97 median_MAPE=9.76%	avg_MAE=13.84 avg_MAPE=12.25% median_MAE=13.78 median_MAPE=12.20%	avg_MAE=7.10 avg_MAPE=10.25% median_MAE=7.10 median_MAPE=10.26%

Fine Tune Result

Model	SBP MAE (mmHg)	DBP MAE (mmHg)
AlexNet	6.20	11.82
ResNet	21.22	11.28
LSTM	13.16	9.4


42
86

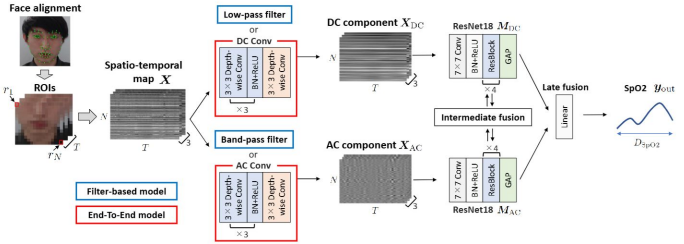
Blood Oxygen / SpO2

Traditional Blood Oxygen Measurement

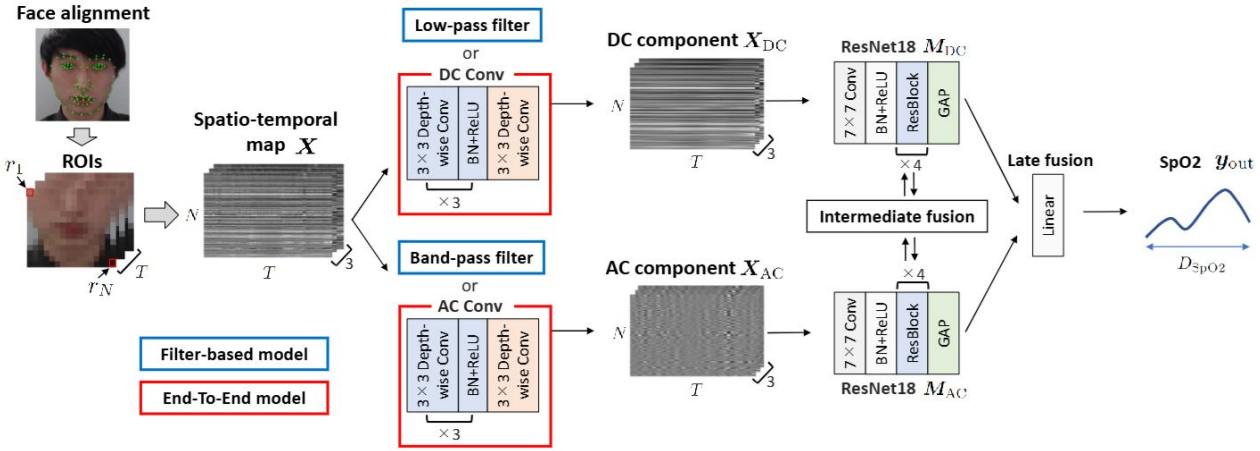


Goal

<1 RMSE & <1 MAE for MCD-rppg dataset

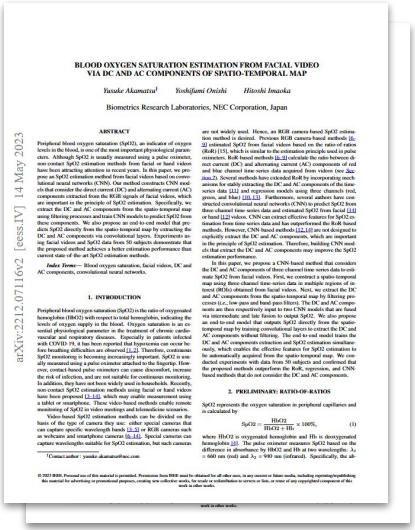


Architecture



Pipeline processes **AC** and **DC component** of **ST map**

2023 Paper, Akamatsu, et. al



BLOOD OXYGEN SATURATION ESTIMATION FROM FACIAL VIDEO VIA DC AND AC COMPONENTS OF SPATIO-TEMPORAL MAP

Yusuke Akamatsu, Yoshifumi Onishi, Hitoshi Imaoka

Biometrics Research Laboratories, NEC Corporation, Japan

ABSTRACT

Peripheral blood oxygen saturation (SpO2), an indicator of oxygen levels in the blood, is one of the most important physiological parameters. Although SpO2 is usually measured using a pulse oximeter, non-contact SpO2 estimation methods from facial or hand videos have been attracting attention in recent years. In this paper, we propose an SpO2 estimation method from facial videos based on convolutional neural networks (CNN). Our method constructs CNN models that consider the direct current (DC) and alternating current (AC) components extracted from the RGB signals of facial videos, which are known to be related to SpO2 values according to theory. We extract the DC and AC components from the spatio-temporal map using filtering processes and train CNN models to predict SpO2 from three components. We also propose an end-to-end model that predicts SpO2 directly from the spatio-temporal map by extracting the DC and AC components via convolutional layers. Experiments using facial videos and SpO2 data from 50 subjects demonstrate that the proposed method achieves a better estimation performance than current state-of-the-art SpO2 estimation methods.

Index Terms— Blood oxygen saturation, facial videos, DC and AC components, convolutional neural networks.

1. INTRODUCTION

Peripheral blood oxygen saturation (SpO2) is the ratio of oxygenated hemoglobin (HbO2) with respect to total hemoglobin, indicating the levels of oxygen supply in the blood. Oxygen saturation is an essential physiological parameter in the treatment of chronic cardiovascular and respiratory diseases. Especially in patients infected with COVID-19, it has been reported that hypoxemia can occur before breathing difficulties are observed [1, 2]. Therefore, continuous SpO2 monitoring is important for these patients. SpO2 is usually measured using a pulse oximeter attached to the fingertip. However, contact-based pulse oximeters can cause discomfort, increase the risk of infection, and are not suitable for continuous monitoring. In addition, they have not been widely used in households. Recently, non-contact SpO2 estimation methods using facial or hand videos have been proposed [3–14], which may enable measurement using a tablet or smartphone. These video-based methods enable remote monitoring of SpO2 in video meetings and telemedicine scenarios.

Video-based SpO2 estimation methods can be divided on the basis of the type of camera they use: either special cameras that can capture specific wavelength bands [3–9] or RGB cameras such as webcams and smartphone cameras [6, 14]. Special cameras can capture wavelengths suitable for SpO2 estimation, but such cameras are not widely used. Hence, an RGB camera based SpO2 estimation method is desired. Previous RGB camera based methods [6, 9] estimated SpO2 from facial videos based on the ratio of ratios (RoR) [15], which is similar to the estimation principle used in pulse oximeters. RoR based methods [6, 9] calculate the ratio between direct current (DC) and alternating current (AC) components of red and blue channel time-series data acquired from videos (see Section 2). Several methods have extended RoR by incorporating mechanisms for stably extracting the DC and AC components of the time-series data [12] and regression models using three channels (red, green, and blue) [10, 13]. Furthermore, several authors have constructed convolutional neural networks (CNN) to predict SpO2 from three-channel time series data and estimated SpO2 from facial [14] or hand [11] videos. These methods have demonstrated that estimation from time series data and has outperformed the RoR based methods. However, CNN based methods [12, 14] are not designed to explicitly extract the DC and AC components, which are important in the principle of SpO2 estimation. Therefore, building CNN models that extract the DC and AC components may improve the SpO2 estimation performance.

In this paper, we propose a CNN based method that considers the DC and AC components of three-channel time series data to estimate SpO2 from facial videos. First, we construct a spatio-temporal map using three channel time series data in multiple regions of interest (ROIs) obtained from facial videos. Next, we extract the DC and AC components from the spatio-temporal map by filtering processes (i.e., low-pass and band-pass filters). The DC and AC components are then respectively input to two CNN models that are based on a two-stream network [16] for video recognition. We also propose an end-to-end model that outputs SpO2 directly from the spatio-temporal map by automatically extracting the DC and AC components without filtering. The end-to-end model trains the DC and AC components extraction and SpO2 estimation simultaneously, which enables the effective features for SpO2 estimation to be automatically acquired from the spatio-temporal map. We conducted experiments with data from 50 subjects and confirmed that the proposed methods outperform the RoR, regression, and CNN based methods that do not consider the DC and AC components.

2. PRELIMINARY: RATIO-OF-RATIOS

SpO2 represents the oxygen saturation in peripheral capillaries and is calculated by

$$SpO2 = \frac{HbO2}{HbO2 + Hb} \times 100\%, \quad (1)$$

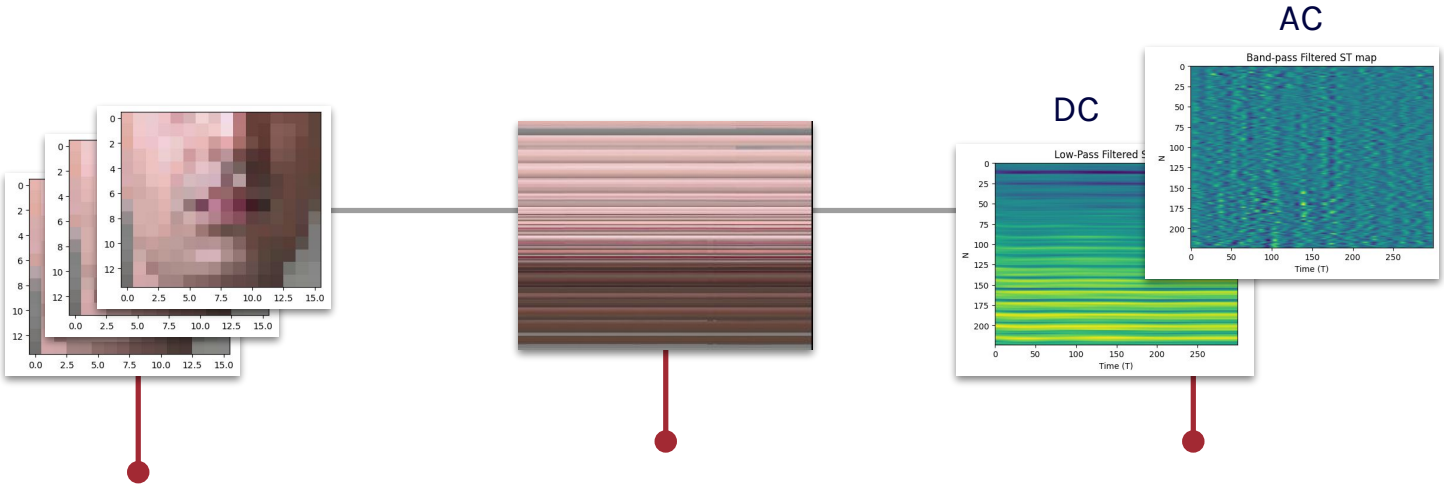
where HbO2 is oxygenated hemoglobin and Hb is deoxygenated hemoglobin [15]. The pulse oximeter measures SpO2 based on the difference in absorbance by HbO2 and Hb at two wavelengths: λ_1 = 660 nm (red) and λ_2 = 940 nm (infrared). Specifically, the ab-

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arXiv:2212.07116v2 [eess.IV] 14 May 2023

Data Processing



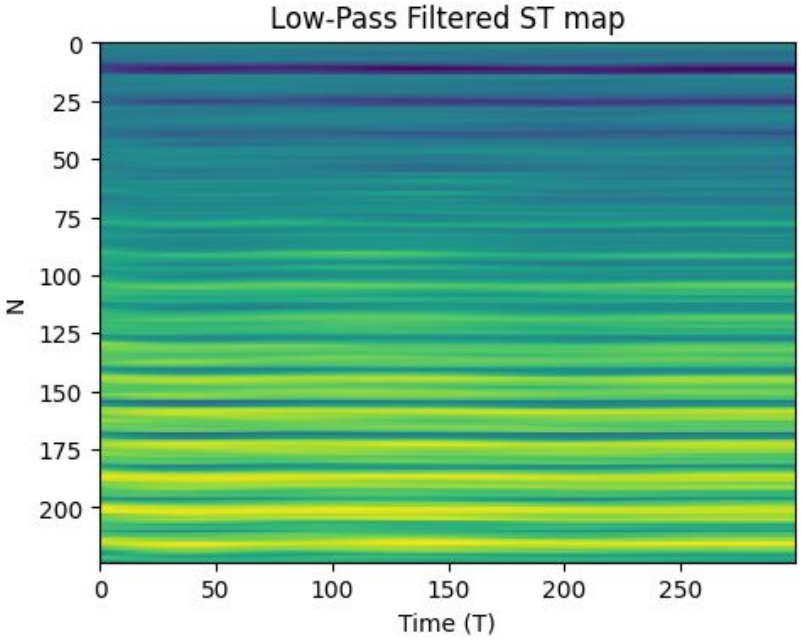
1. Extract ROI

2. Spatio-temporal map

3. Filter for AC/DC maps

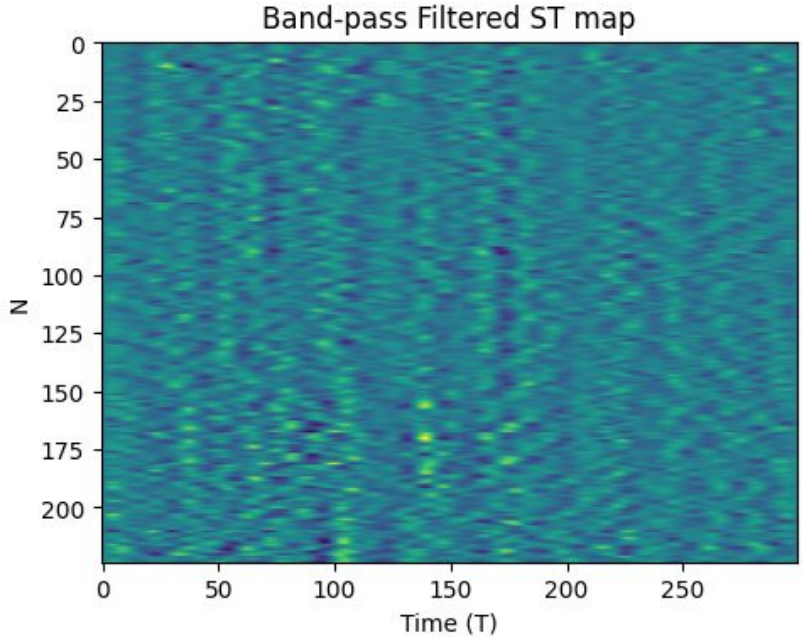
SpO_2 [4/8]

DC Signal



Baseline signal

AC Signal



Changing signal relating to blood movement

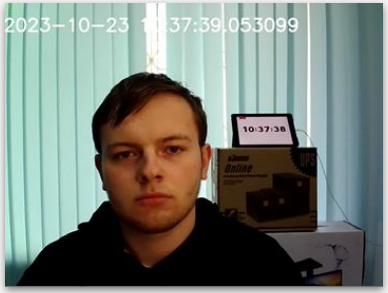
SpO_2 [5/8]

Accuracy

0.705
RMSE

0.705
MAE

Sample



Video

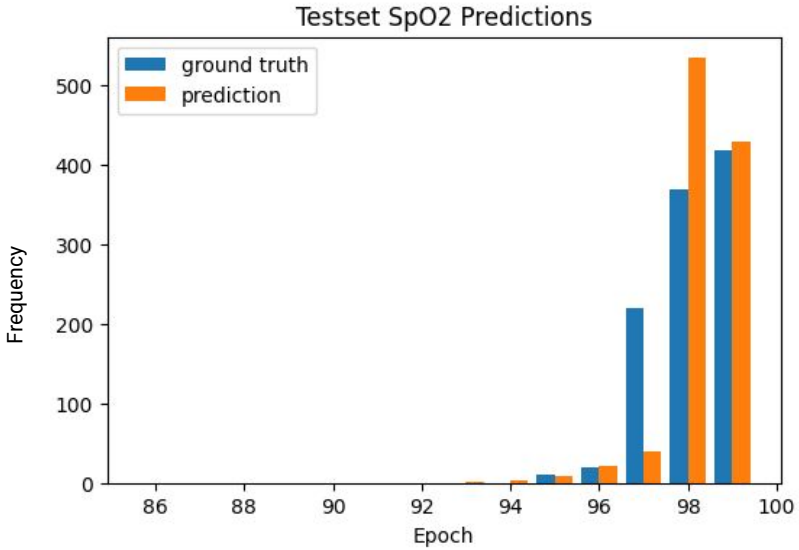


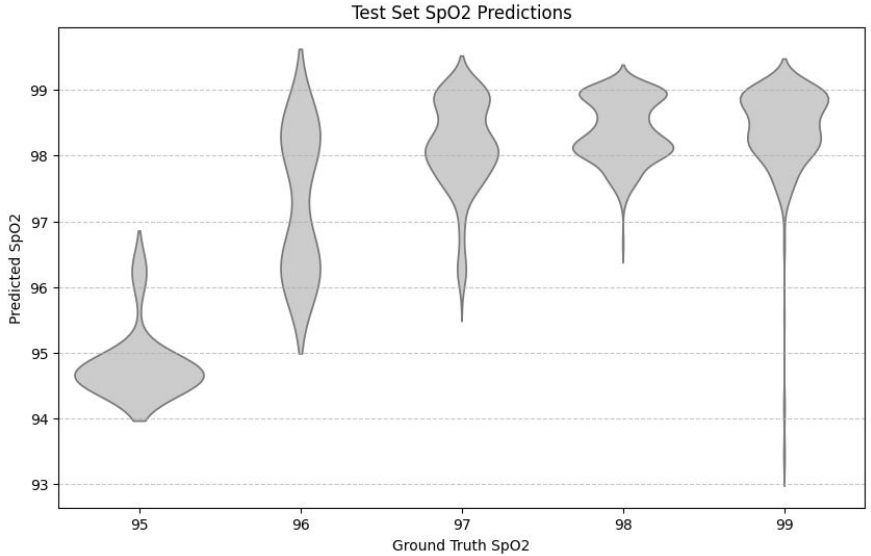
ST-Map

97.9
Prediction

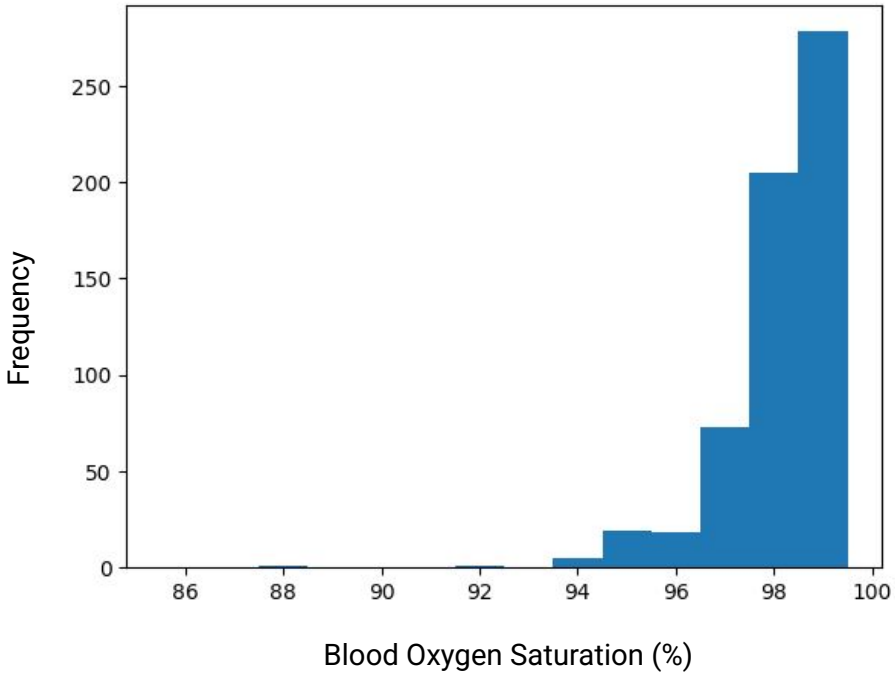
97
Ground Truth

Weak Generalization





Dataset Limitations

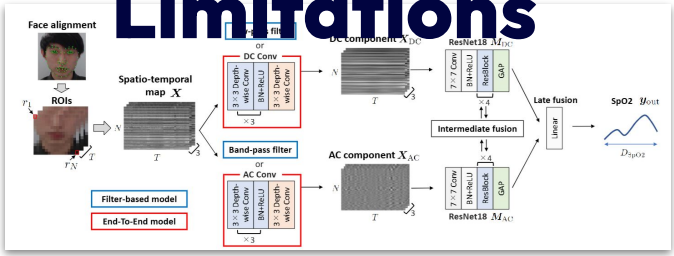


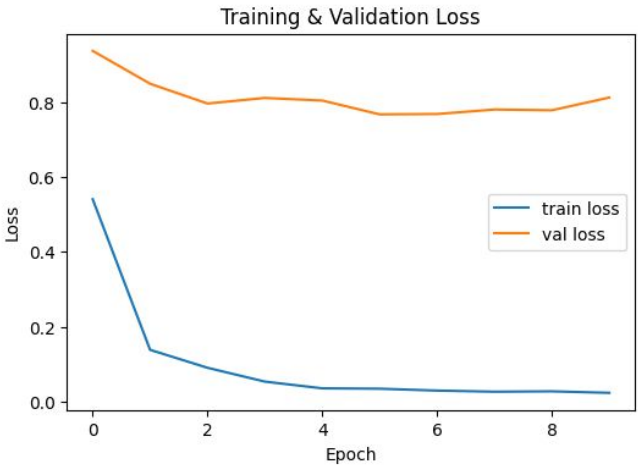
0.875
RMSE

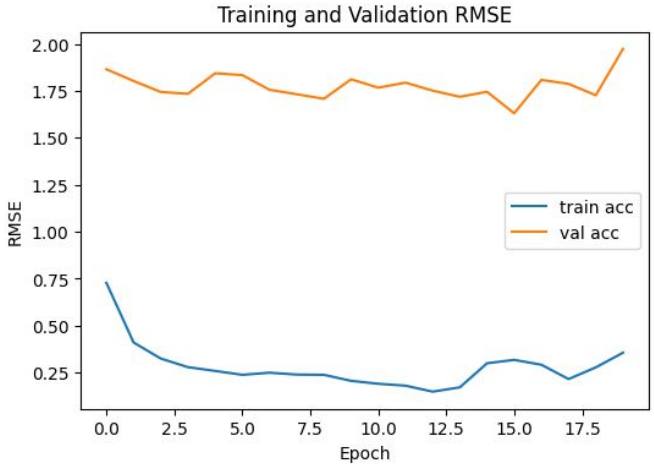
0.700
MAE

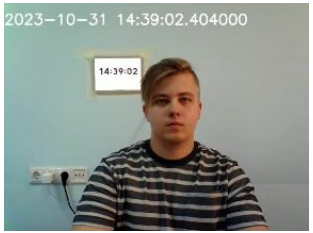
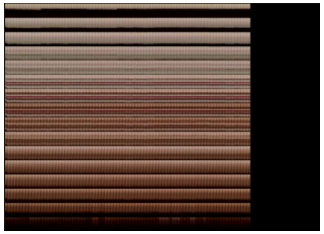
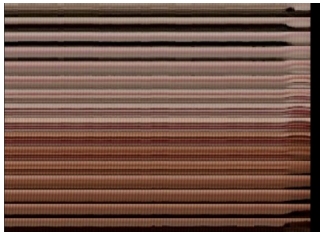
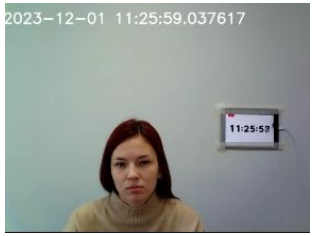
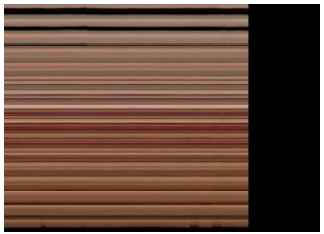
Blind guess (Test dataset)

Architecture Limitations







Video	ST-Map	Prediction	Ground Truth
		98.46	98.0
		98.08	99.0
		98.72	99.0




42
86

Future Obstacle: Data Pattern before and after Exercise



1020_FullHDwebcam_before

5s

+

SpO_2 drop, Blood Pressure increase.



1020_FullHDwebcam_after

5s

BMI

Body Mass Index (BMI = weight[kg]/height ^{2}[m^2])

Frame Segments and Face Detection → Feature Extraction and Regression → BMI



.avi to .jpg(224 x 224)

Pretrained CNN Models

VGG19

ResNet50

Group	Range
Underweight	BMI < 18.5
Healthy/Normal	18.5 ≤ BMI <25
Overweight	25 ≤ BMI< 30
Obesity	BMI > 30

Body Mass Index (BMI = weight[kg]/height ^{2}[m^2])

Frame Segments and Face Detection → Feature Extraction and Regression → BMI



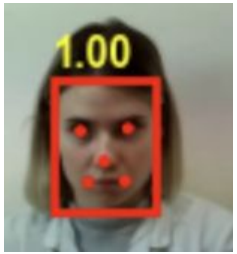
<18.5	18.5–24.9	25–29.9	30-39.9	40<
Underweight	Normal Weight	Class I Obesity	Class II Obesity	Class III Obesity

Frame Segments and Face Detection



1020_FullHDwebcam_before

MTCNN





BMI Pipeline

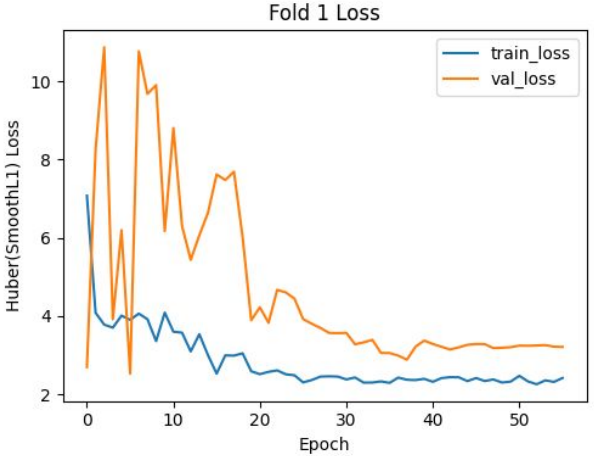


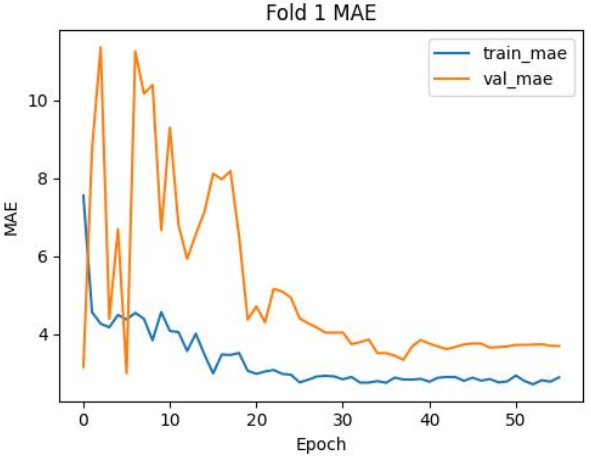
Residual Blocks Regression

Dense + BN + ReLU + Dropout blocks with **shortcuts**

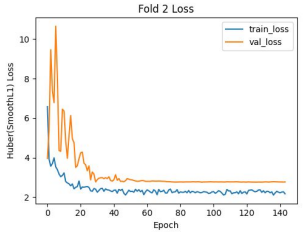
BMI Pipeline

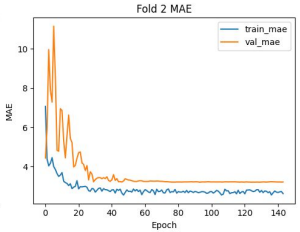
avg MAE: 3.07

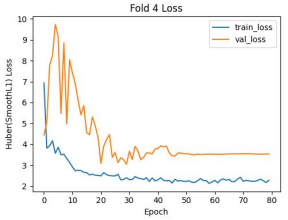


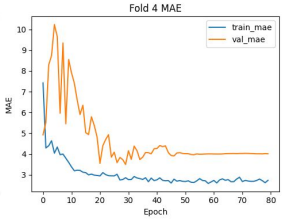


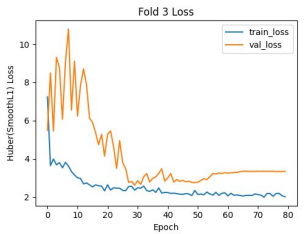
Dataset

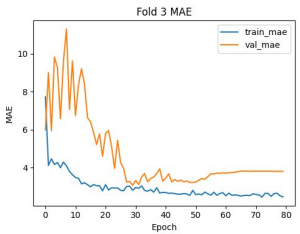


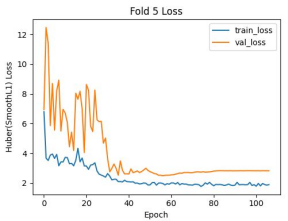


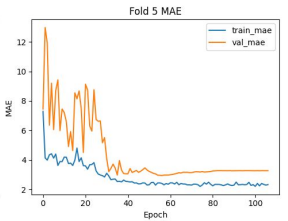












Feature Extraction and Regression



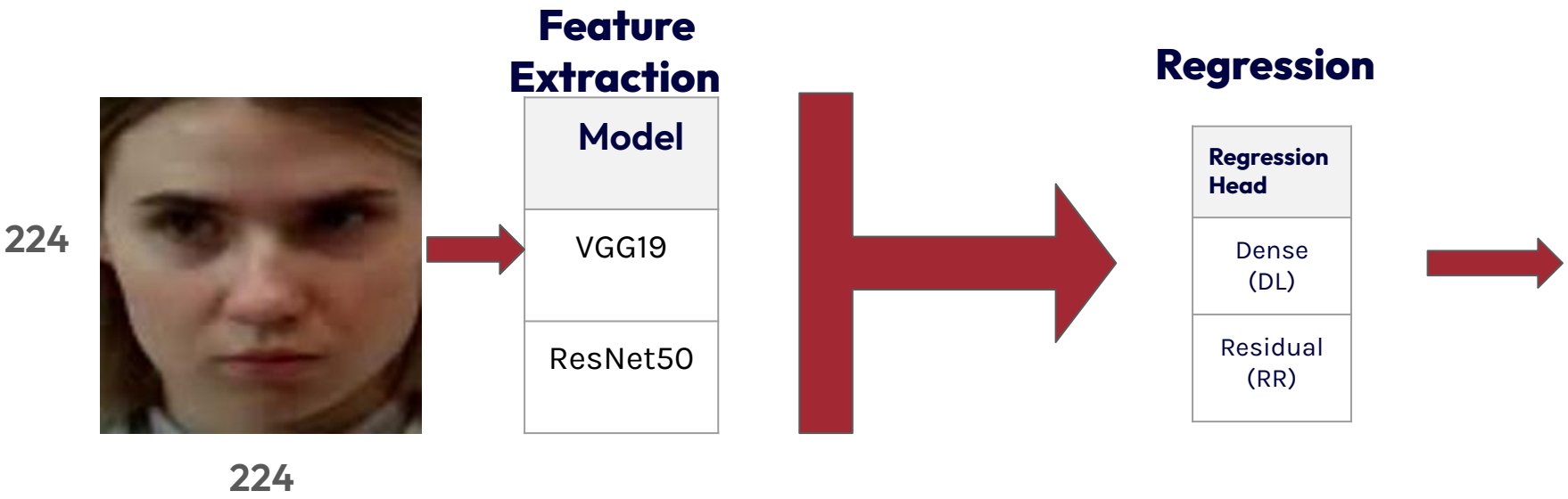
Feature Extraction

Model	Strength	Limitation
VGG19	Simple, Stable	Heavy, Old
ResNet50	Deep, Efficient	Complex, Sensitive

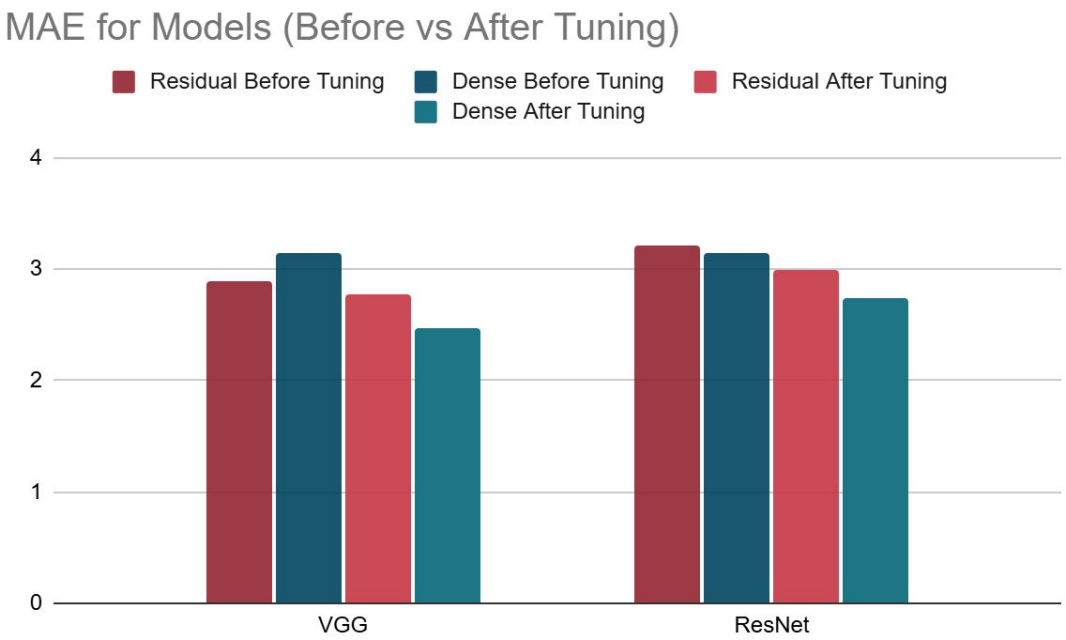
Regression

Regression Head	Strength	Limitation
Dense (DL)	Stable, Low-overfit	Limited, Underfits
Residual (RR)	Powerful, Expressive	Overfits, Heavy

Feature Extraction and Regression



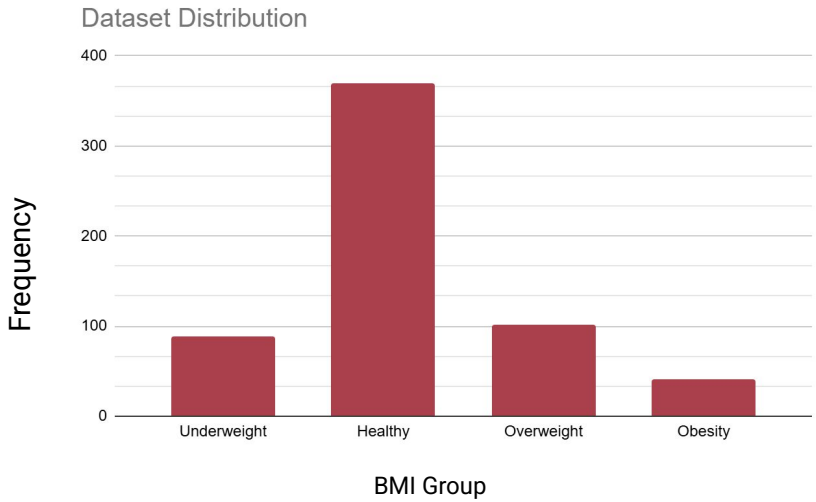
Model Comparison



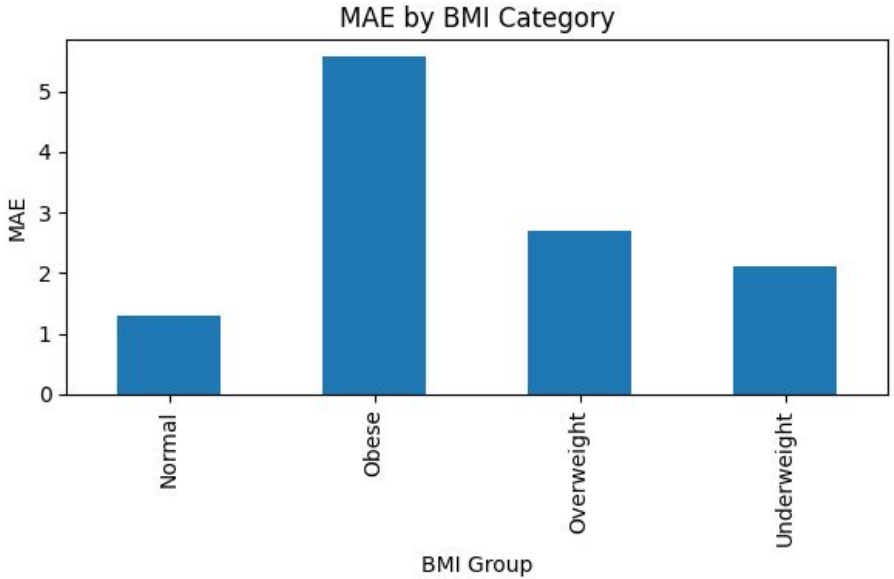
2.47
VGG19 + Dense

Lowest MAE

Dataset Limitations and Impact



Unbalanced Dataset



MAE for Each BMI Group

Conclusion

3 Biomarker Predictions



Body Mass Index

Blood Oxygen

Blood Pressure

23

98%

115 sbp/75 dbp

Practical Value









Conclusion

All you need is syd

Life Quality AI

